From market states to robust portfolios: adaptive semi-infinite CVaR optimization with real financial data

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Abstract

Portfolio risk depends heavily on prevailing macroeconomic conditions, yet standard robust portfolio optimization models frequently represent those conditions through a rigid, discrete set of regimes or an unconditional ambiguity set. We introduce a data-driven semi-infinite portfolio optimization framework in which each point of a continuous market-state space induces a conditional empirical return distribution. Market states are characterized by observable financial indicators, specifically the implied volatility index (VIX) and the equity market drawdown. Non-parametric multivariate kernel weights connect historical observations to any continuous target state, avoiding arbitrary discrete regime segmentation. Portfolio risk is measured by the worst-case conditional value-at-risk (CVaR) over a compact continuous state space. To solve the resulting semi-infinite program efficiently, we develop an adaptive exchange algorithm that alternates between a finite linear programming master problem and a continuous-state oracle, iteratively identifying binding stress states. We establish theoretical existence, continuity, and convergence properties of the model, and we analyze the approximation guarantees when using inexact continuous separation oracles. We conduct an extensive rolling-window empirical study over three decades of US industry portfolio data (1990 to 2026). The continuous robust model is systematically benchmarked against naive diversification (1/N), minimum variance, unconditional nominal CVaR, and a four-quadrant finite-regime CVaR model. The empirical evaluation rigorously examines realized downside risk, effective asset concentration, temporal turnover, transaction cost drag, pre-defined crisis periods, computational scalability relative to dense grids, and studentized circular block-bootstrap hypothesis testing on Sharpe ratio differences. The continuous-state robust framework provides enhanced capital preservation during acute market dislocations while maintaining computational tractability for practical asset management.

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1 Introduction and motivation

The classical mean-variance framework introduced by Markowitz [1952] established the quantitative foundation of modern portfolio theory. However, the practical implementation of mean-variance optimization has long been hindered by its extreme sensitivity to parameter estimation errors. Small perturbations in expected returns or covariance estimates often produce erratic allocations, excessive portfolio turnover, and severe out-of-sample disappointment. To address these vulnerabilities, researchers have developed robust portfolio optimization techniques that optimize allocations against the worst-case parameters within prespecified uncertainty sets [Goldfarb and Iyengar, 2003, Fabozzi et al., 2007].

Traditional robust portfolio models predominantly focus on unconditional uncertainty sets constructed around historical sample moments or empirical distributions [Delage and Ye, 2010, Mohajerin Esfahani and Kuhn, 2018]. While these models effectively mitigate sensitivity to estimation noise, they typically treat market dynamics as stationary and homogeneous across time. In reality, financial markets exhibit pronounced state-dependent behavior. During tranquil market environments, asset correlations remain moderate and return dispersions offer meaningful diversification benefits. In contrast, during periods of acute financial distress, volatility spikes sharply, asset correlations converge toward unity, and tail risks materialize simultaneously across sectors.

To incorporate state-dependency, a common practice in financial econometrics is to employ discrete regime-switching models [Ang and Bekaert, 2002]. These models segment market history into a small, finite collection of latent states, such as low-volatility expansionary regimes and high-volatility contractionary regimes. While finite regime classifications offer conceptual simplicity and computational tractability, they impose an artificial discretization on what is fundamentally a continuous economic reality. Financial markets do not transition abruptly between a few discrete states. Instead, market stress evolves along a continuous spectrum of macroeconomic and financial conditions. A sudden jump between two discrete regimes fails to capture intermediate stress levels and ignores boundary dynamics where vulnerability to systemic shocks varies continuously.

In this paper, we propose a continuous-state robust portfolio optimization framework that bridges the gap between non-parametric conditional risk modeling and semi-infinite optimization. Rather than partitioning market history into discrete regimes, we define a continuous, compact state space $\mathcal{U} \subset \mathbb{R}^2$ spanned by observable financial indicators: the CBOE Volatility Index (VIX) and the equity market drawdown. Every point $\theta \in \mathcal{U}$ represents a distinct market environment and induces a unique conditional empirical return distribution constructed via multivariate Nadaraya-Watson kernel weighting.

The investor seeks an allocation that minimizes the worst-case Conditional Value-at-Risk (CVaR) over the entire continuous state space $\mathcal{U}$. Because the state space is uncountable, this formulation gives rise to a Semi-Infinite Program (SIP) with an infinite number of joint risk constraints. Direct solution via dense spatial discretization becomes computationally prohibitive and scales poorly in high dimensions. To solve the continuous-state SIP efficiently, we design an adaptive exchange algorithm. The method alternates between solving a finite master linear program over a sparse active subset of stress states and executing a continuous global oracle that searches for the most adverse state for the current candidate portfolio.

This study makes several methodological, theoretical, and empirical contributions to the literature on robust asset allocation:

First, we formalize the continuous-state robust CVaR problem with a mathematically complete master epigraph formulation and a rigorously defined compact state space. We position this approach relative to recent advances in conditional stochastic optimization and distributionally robust optimization with side information [Bertsimas and Kallus, 2020, Esteban-Pérez and Morales, 2022, Nguyen et al., 2021].

Second, we provide theoretical foundations for the continuous-state framework, establishing the compactness of the decision and state spaces, the continuity of the kernel weighting mapping, the joint continuity and convexity of the conditional CVaR objective, and the existence of an optimal robust allocation. We also formalize the theoretical properties of the exchange algorithm under exact and inexact continuous separation oracles, characterizing the empirical optimality gap when global search heuristics are employed [Oustry and Cerulli, 2025].

Third, we implement a full experimental backtest over a 30-year span (1990 to 2026) using daily returns from the Kenneth French 30 Industry Portfolios alongside contemporaneous VIX and drawdown data. We compare the continuous robust portfolio against four benchmarks: the naive 1/N allocation [DeMiguel et al., 2009], the global minimum variance portfolio, the unconditional nominal CVaR portfolio, and a four-quadrant finite-regime CVaR portfolio.

Fourth, we conduct a comprehensive empirical evaluation reporting 14 distinct performance metrics, gross and net of transaction costs. We analyze pre-defined historical crisis windows (the Dot-Com crash, the 2008 Global Financial Crisis, the COVID-19 shock, and the 2022 inflation tightening), perform sensitivity sweeps over transaction costs and kernel bandwidths, evaluate computational convergence against dense grid solutions, and conduct studentized circular block-bootstrap hypothesis testing following Ledoit and Wolf [2008].

2 Related work and literature positioning

Our methodology intersects three active streams of quantitative research: robust portfolio selection, conditional stochastic programming with side information, and semi-infinite optimization algorithms.

2.1 Robust and distributionally robust portfolio optimization

Classical robust portfolio selection addresses the sensitivity of Markowitz mean-variance optimization to estimation noise by assuming that expected return vectors and covariance matrices reside in deterministic uncertainty sets [Goldfarb and Iyengar, 2003, Fabozzi et al., 2007]. These formulations yield second-order cone or semidefinite programs that provide worst-case guarantees. However, moment-based robust models frequently rely on implicit symmetry assumptions that fail to reflect the severe skewness, heavy tails, and asymmetric dependence structures inherent in financial return distributions.

To capture tail risk directly, Rockafellar and Uryasev [2000, 2002] introduced the Conditional Value-at-Risk (CVaR) as a coherent risk measure that can be optimized efficiently via linear programming. Building on this foundation, distributionally robust optimization (DRO) seeks to minimize risk over an ambiguity set of probability distributions constructed around an empirical baseline. Modern DRO frameworks define ambiguity balls using moment constraints [Delage and Ye, 2010] or optimal transport metrics such as the Wasserstein distance [Mohajerin Esfahani and Kuhn, 2018]. While DRO provides rigorous out-of-sample guarantees against statistical sample perturbations, standard DRO models remain unconditional, assuming a single stationary distribution across all market regimes.

2.2 Conditional stochastic optimization and side information

A rapidly expanding body of literature investigates how observable side information (covariates) can be integrated into data-driven decision-making. Bertsimas and Kallus [2020] established a general framework for predictive prescriptions, demonstrating that non-parametric kernel regression, k-nearest neighbors, and tree-based weights can effectively condition decision problems on prevailing covariates. In the context of risk management, Scaillet [2004] examined non-parametric kernel estimation of conditional expected shortfall.

More recently, researchers have unified side information with robust optimization. Esteban-Pérez and Morales [2022] developed distributionally robust stochastic programs with side information based on trimmed Wasserstein balls, applying their method to conditional energy and portfolio problems. Nguyen et al. [2021] formulated robust conditional portfolio decisions via optimal transport metrics centered on covariate-conditioned empirical distributions.

Our framework differs from conditional DRO in its specific mathematical construction. In conditional DRO, the decision-maker constructs an ambiguity set around the conditional distribution corresponding to the currently observed state y_t. In our continuous-state robust model, we do not restrict attention to a single observed state. Instead, we seek a portfolio that remains robust across the entire family of conditional distributions induced by any plausible state $\theta \in \mathcal{U}$. We refer to this approach as continuous-state robust optimization over kernel-weighted empirical conditional distributions.

2.3 Semi-infinite programming and exchange algorithms

Semi-infinite programming (SIP) considers optimization problems with a finite number of decision variables and an infinite index set of constraints [Hettich and Kortanek, 1993, López and Still, 2007]. SIP techniques have found applications in engineering design, robust control, and spectral estimation, but their operational use in financial portfolio allocation has been constrained by computational complexity.

The standard solution paradigm for convex SIPs is the cutting-plane or exchange algorithm. At each iteration, the master problem solves a relaxed subproblem over a finite subset of constraints, while a separation oracle identifies the most violated constraint across the continuous index set. When the oracle can be solved to certified global optimality, the exchange algorithm generates monotonically non-decreasing lower bounds and certified upper bounds that converge to the true global optimum. In practical high-dimensional settings where the oracle is non-convex, exact separation is challenging. Oustry and Cerulli [2025] formalized the convergence and error propagation of convex SIP algorithms equipped with inexact separation oracles, establishing explicit bounds on infeasibility and suboptimality. We build upon these principles to provide theoretical grounding for our adaptive exchange method.

3 Continuous market-state modeling

We consider an investment universe consisting of N risky financial assets over a discrete time horizon $t = 1, \ldots, T$. Let $x_t = (x_{1,t}, \ldots, x_{N,t})^\top \in \mathbb{R}^N$ denote the vector of realized asset returns from trading day $t-1$ to day t. We hypothesize that the joint conditional distribution of asset returns is modulated by prevailing macroeconomic and market state variables observable at time $t-1$, denoted by the vector $y_{t-1} \in \mathbb{R}^M$.

3.1 State variable selection and normalization

In this study, we focus on two primary financial indicators that capture distinct dimensions of market risk ($M = 2$):

1. **Implied Market Volatility (V_t)**: The Chicago Board Options Exchange (CBOE) Volatility Index (VIX), which reflects forward-looking market uncertainty and implied option volatility.
2. **Market Drawdown (D_t)**: The peak-to-trough decline of the broad equity market index over a trailing quarterly lookback window of $L = 63$ trading days, defined as:

$$D_t = 1 - \frac{P_t}{\max_{s \in [t-L,t]} P_s} \tag{1}$$

   where P_t is the cumulative total return index of the broad market.

The state vector at time $t-1$ is $y_{t-1} = (V_{t-1}, D_{t-1})^\top$. Because implied volatility and percentage drawdowns have different numerical scales and variance structures, we standardize the state variables over the historical estimation window. Let $\bar{y} \in \mathbb{R}^2$ and $\hat{\Sigma}_y \in \mathbb{R}^{2\times 2}$ denote the sample mean vector and sample covariance matrix of the state variables over the training sample. The standardized historical state vectors are given by $\tilde{y}_{t-1} = \hat{\Sigma}_y^{-1/2}(y_{t-1} - \bar{y})$.

3.2 Definition of the continuous state space

We define the continuous state space $\mathcal{U} \subset \mathbb{R}^2$ as a compact bounding rectangle containing all historically observed market states, augmented by an outer safety margin $\delta > 0$ to account for potential unobserved stress levels:

$$\mathcal{U} = [v_{\min} - \delta_v, v_{\max} + \delta_v] \times [d_{\min} - \delta_d, d_{\max} + \delta_d] \tag{2}$$

where $v_{\min} = \min_t V_{t-1}$, $v_{\max} = \max_t V_{t-1}$, $d_{\min} = \min_t D_{t-1} = 0$, and $d_{\max} = \max_t D_{t-1}$. In our implementation, we set $\delta_v = 0.10(v_{\max} - v_{\min})$ and $\delta_d = 0.10(d_{\max} - d_{\min})$. By construction, $\mathcal{U}$ is a closed and bounded subset of $\mathbb{R}^2$, ensuring compactness.

3.3 Kernel-weighted conditional empirical distributions

For any continuous state query $\theta = (v_\theta, d_\theta)^\top \in \mathcal{U}$, we construct a state-dependent empirical probability distribution over the historical return realizations $\{x_1, \ldots, x_T\}$. Each historical return observation x_t is assigned a Nadaraya-Watson kernel weight based on the proximity of its preceding state y_{t-1} to the target state θ:

$$p_t(\theta) = \frac{K_H(y_{t-1} - \theta)}{\sum_{s=1}^{T} K_H(y_{s-1} - \theta)} \tag{3}$$

where $K_H(u) = |H|^{-1/2}(2\pi)^{-M/2} \exp\left(-\frac{1}{2}u^\top H^{-1} u\right)$ is a multivariate Gaussian kernel with symmetric positive-definite bandwidth matrix $H \in \mathbb{R}^{2\times 2}$.

We specify the bandwidth matrix using the multivariate rule-of-thumb [Silverman, 1986]:

$$H = \left(\frac{4}{M+2}\right)^{\frac{2}{M+4}} T^{-\frac{2}{M+4}} \hat{\Sigma}_y = T^{-1/3}\hat{\Sigma}_y \tag{4}$$

for $M = 2$. This choice balances bias and variance while naturally adapting to the empirical covariance structure of the state variables.

Because the Gaussian kernel is strictly positive everywhere on $\mathbb{R}^2$, the denominator $\sum_{s=1}^{T} K_H(y_{s-1} - \theta)$ is strictly positive for all $\theta \in \mathcal{U}$. Consequently, the kernel weights satisfy $p_t(\theta) > 0$ and $\sum_{t=1}^{T} p_t(\theta) = 1$ for every $\theta \in \mathcal{U}$. Each continuous state θ thus induces a well-defined conditional empirical distribution:

$$\mathbb{P}(X|\theta) = \sum_{t=1}^{T} p_t(\theta)\delta_{x_t} \tag{5}$$

where δ_{x_t} denotes the Dirac delta mass at historical return vector x_t.

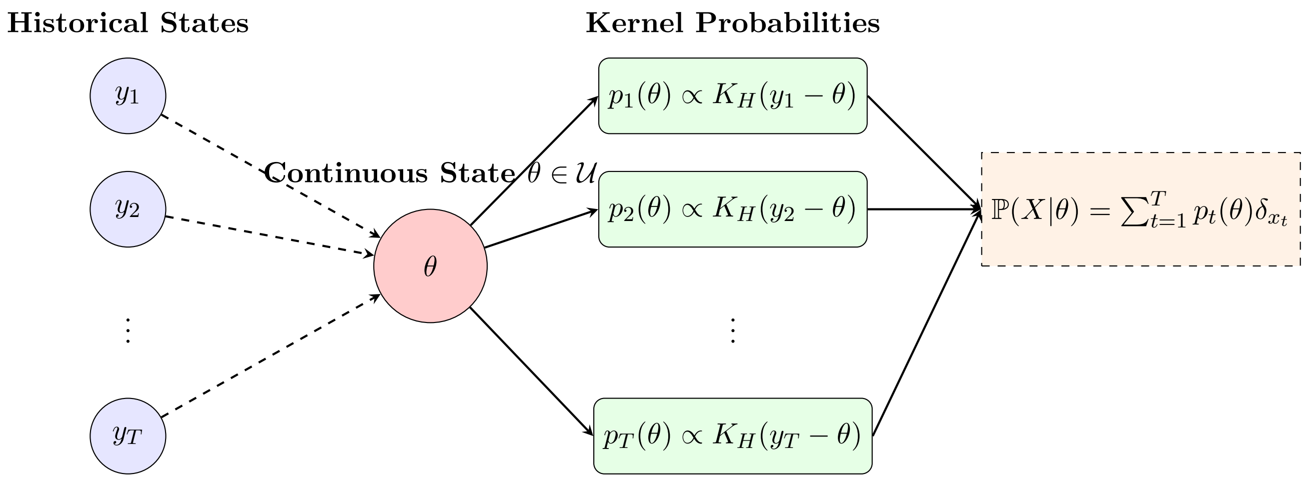


Figure 1: Schema 1: Continuous market state mapping using multivariate kernel density weighting to construct conditional empirical return distributions from observable financial indicators.

3.4 Effective sample size diagnostics

When evaluating conditional risk in regions of the state space with low historical data density, the kernel estimator concentrates probability mass on a small subset of observations. To quantify local data richness, we compute the Effective Sample Size (ESS) for any state $\theta \in \mathcal{U}$:

$$\mathrm{ESS}(\theta) = \frac{1}{\sum_{t=1}^{T} p_t(\theta)^2} \tag{6}$$

When weights are uniformly distributed ($p_t(\theta) = 1/T$), the ESS reaches its theoretical maximum $\mathrm{ESS}(\theta) = T$. Conversely, if all weight concentrates on a single historical observation, $\mathrm{ESS}(\theta) = 1$. In our empirical analysis, we track the ESS of all active stress states identified by the optimization algorithm to ensure that the robust solutions do not degenerate into single-scenario overfits.

4 Continuous-state robust CVaR optimization

Let $w = (w_1, \dots, w_N)^\top \in \mathbb{R}^N$ denote the portfolio weight vector. The portfolio loss under realized return vector x_t is $-x_t^\top w$. We define the feasible portfolio set $W \subset \mathbb{R}^N$ under standard institutional long-only constraints and a target expected return constraint:

$$W = \left\{ w \in \mathbb{R}^N : \sum_{i=1}^{N} w_i = 1, \quad w_i \geq 0 \quad \forall i = 1, \dots, N, \quad \hat{\mu}^\top w \geq \mu_{\text{target}} \right\} \tag{7}$$

where $\hat{\mu} = \frac{1}{T} \sum_{t=1}^{T} x_t$ is the unconditional sample mean return vector, and μ_{target} is the required expected return. The set W is non-empty whenever $\mu_{\text{target}} \leq \max_i \hat{\mu}_i$, and it is closed and bounded, hence compact.

4.1 Conditional value-at-risk formulation

For a given portfolio allocation $w \in W$ and a fixed state $\theta \in \mathcal{U}$, we measure portfolio downside risk using the Conditional Value-at-Risk at tail probability level $\tau \in (0, 1)$, corresponding to confidence level $\alpha = 1 - \tau$. We set $\tau = 0.05$ ($\alpha = 0.95$) throughout the empirical analysis. Following the variational representation of Rockafellar and Uryasev [2000], the conditional CVaR evaluated at state θ is:

$$\Phi_\tau(w, \theta) = \min_{z \in \mathbb{R}} \left\{ z + \frac{1}{\tau} \sum_{t=1}^{T} p_t(\theta) [-x_t^\top w - z]_+ \right\} \tag{8}$$

where $z \in \mathbb{R}$ represents the conditional Value-at-Risk (VaR) at level α, and $[u]_+ = \max(0, u)$.

4.2 The semi-infinite programming formulation

The continuous-state robust portfolio optimization problem seeks an allocation $w \in W$ that minimizes the worst-case conditional CVaR across all market states in the compact domain $\mathcal{U}$:

$$v^* = \min_{w \in W} \sup_{\theta \in \mathcal{U}} \Phi_\tau(w, \theta) \tag{9}$$

Introducing an auxiliary epigraph scalar $\eta \in \mathbb{R}$ representing the robust risk level, the problem is equivalently formulated as the following convex Semi-Infinite Program (SIP):

$$\min_{w\in\mathbb{R}^N,\eta\in\mathbb{R}} \quad \eta \tag{10}$$

$$\text{s.t.} \quad \Phi_\tau(w,\theta) \le \eta \quad \forall\theta \in \mathcal{U} \tag{11}$$

$$\sum_{i=1}^{N} w_i = 1 \tag{12}$$

$$w_i \ge 0 \quad \forall i = 1,\ldots,N \tag{13}$$

$$\hat{\mu}^\top w \ge \mu_{\text{target}} \tag{14}$$

Because $\mathcal{U}$ is an uncountably infinite set, the problem contains an infinite family of continuous convex constraints $\Phi_\tau(w,\theta) \le \eta$.

4.3 Theoretical properties: existence, continuity, and convexity

We now establish the mathematical foundations ensuring the well-posedness of the continuous-state robust CVaR problem.

Proposition 1 (Compactness and continuity)**.** *Assume that the bandwidth matrix H is symmetric positive-definite and the target expected return μ_{target} is feasible. Then:*

1. *The feasible portfolio set W and the market state space $\mathcal{U}$ are non-empty, compact subsets of $\mathbb{R}^N$ and $\mathbb{R}^2$, respectively.*
2. *For every $t = 1,\ldots,T$, the kernel weight function $\theta \mapsto p_t(\theta)$ is infinitely differentiable and strictly positive on $\mathcal{U}$.*
3. *The conditional CVaR function $(w,\theta) \mapsto \Phi_\tau(w,\theta)$ is jointly continuous on $W \times \mathcal{U}$.*
4. *For every fixed $\theta \in \mathcal{U}$, the function $w \mapsto \Phi_\tau(w,\theta)$ is real-valued and convex on W.*

Proof. Item 1 follows directly from the definition of the standard unit simplex intersected with a half-space and the closed bounding box construction of $\mathcal{U}$. For Item 2, the Gaussian kernel $K_H(u)$ is smooth and strictly positive everywhere. The denominator $\sum_{s=1}^{T} K_H(y_{s-1}-\theta) \ge \min_{\theta\in\mathcal{U}} K_H(y_0-\theta) > 0$ is bounded away from zero on compact $\mathcal{U}$. Thus $p_t(\theta)$ is a ratio of smooth, positive functions with a non-vanishing denominator, making it smooth and strictly positive. For Item 3, for each fixed t, the loss function $(w,z) \mapsto [-x_t^\top w - z]_+$ is jointly continuous and convex. Because $p_t(\theta)$ is continuous, the objective function $F(w,z,\theta) = z + \frac{1}{\tau}\sum_{t=1}^{T} p_t(\theta)[-x_t^\top w - z]_+$ is jointly continuous on $\mathbb{R}^N \times \mathbb{R} \times \mathcal{U}$. The minimization over z occurs over a compact interval because the optimal z is bounded by the minimum and maximum possible portfolio losses over the compact set W. By Berge's Maximum Theorem, the marginal function $\Phi_\tau(w,\theta) = \min_z F(w,z,\theta)$ is continuous on $W \times \mathcal{U}$. For Item 4, for fixed θ, $\Phi_\tau(\cdot,\theta)$ is the partial minimization over z of the jointly convex function $(w,z) \mapsto F(w,z,\theta)$, which preserves convexity in w. $\square$

Theorem 1 (Existence of optimal robust allocation)**.** *Under the assumptions of Proposition 1, the robust objective function:*

$$G(w) = \sup_{\theta \in \mathcal{U}} \Phi_\tau(w, \theta) \tag{15}$$

is continuous and convex on W. Furthermore, the supremum is attained for every $w \in W$, and the continuous-state robust portfolio optimization problem admits an optimal solution $w^* \in W$.

Proof. Because $\mathcal{U}$ is compact and $\Phi_\tau(w, \cdot)$ is continuous on $\mathcal{U}$ for every $w \in W$, the Extreme Value Theorem guarantees that the supremum is achieved at some $\theta^*(w) \in \mathcal{U}$, so $G(w) = \max_{\theta \in \mathcal{U}} \Phi_\tau(w, \theta)$. As the pointwise maximum of a family of convex functions $\{w \mapsto \Phi_\tau(w, \theta)\}_{\theta \in \mathcal{U}}$, $G(w)$ is convex on W. Joint continuity of $\Phi_\tau(w, \theta)$ on compact $W \times \mathcal{U}$ implies by Berge's Maximum Theorem that $G(w)$ is continuous on W. Finally, minimizing the continuous function $G(w)$ over the non-empty compact set W guarantees the existence of an optimal solution $w^* \in W$ by the Weierstrass Theorem. □

5 The adaptive exchange algorithm

Direct discretization of the continuous state space $\mathcal{U}$ over a fine Cartesian grid $\widehat{\mathcal{U}} = \{\theta_1, \ldots, \theta_K\}$ yields a large-scale linear program with $K \times T$ auxiliary variables and constraints. For realistic rolling backtests requiring hundreds of sequential optimizations, dense grid solvers suffer from excessive memory footprints and slow convergence. We design an adaptive exchange algorithm that iteratively generates only the binding stress states.

5.1 The finite master linear program

At iteration $k \geq 1$, the master problem optimizes portfolio weights w and the robust risk level η subject to a finite active subset of market states $\mathcal{U}_k = \{\theta^{(1)}, \ldots, \theta^{(m_k)}\} \subset \mathcal{U}$. For each active state $\theta^{(j)} \in \mathcal{U}_k$, we introduce a state-specific VaR variable $z_{\theta^{(j)}} \in \mathbb{R}$ and scenario excess loss variables $u_{t,\theta^{(j)}} \geq 0$. The master problem is the following finite Linear Program (LP):

$$\mathrm{LB}_k = \min_{w,\eta,\{z_\theta\},\{u_{t,\theta}\}} \quad \eta \tag{16}$$

$$\text{s.t.} \quad z_\theta + \frac{1}{\tau} \sum_{t=1}^{T} p_t(\theta) u_{t,\theta} \leq \eta \quad \forall \theta \in \mathcal{U}_k \tag{17}$$

$$u_{t,\theta} \geq -x_t^\top w - z_\theta \quad \forall t = 1, \ldots, T, \forall \theta \in \mathcal{U}_k \tag{18}$$

$$u_{t,\theta} \geq 0 \quad \forall t = 1, \ldots, T, \forall \theta \in \mathcal{U}_k \tag{19}$$

$$\sum_{i=1}^{N} w_i = 1, \quad w_i \geq 0 \quad \forall i = 1, \ldots, N \tag{20}$$

$$\hat{\mu}^\top w \geq \mu_{\text{target}} \tag{21}$$

Solving the master LP yields a candidate allocation w_k and a lower bound LB_k on the true optimal value v^*, because $\mathcal{U}_k \subset \mathcal{U}$ represents a relaxed constraint set.

5.2 The continuous separation oracle

Given the candidate allocation w_k, the separation oracle searches the continuous state space $\mathcal{U}$ to identify the most severe market condition that maximizes conditional CVaR:

$$\theta^* = \arg\max_{\theta \in \mathcal{U}} \Phi_\tau(w_k, \theta) \tag{22}$$

and records the corresponding upper bound $\mathrm{UB}_k = \Phi_\tau(w_k, \theta^*)$.

The conditional CVaR evaluation $\Phi_\tau(w_k, \theta)$ requires solving a one-dimensional convex optimization problem over $z \in \mathbb{R}$. For a fixed state θ, the objective $z \mapsto z + \frac{1}{\tau}\sum_{t=1}^{T} p_t(\theta)[-x_t^\top w_k - z]_+$ is piecewise linear and convex. Its global minimum is attained at the weighted empirical quantile $z^*(\theta) = \mathrm{VaR}_\tau(w_k, \theta)$, which can be computed in $O(T \log T)$ time by sorting the portfolio losses $l_t = -x_t^\top w_k$ and finding the smallest loss whose cumulative kernel weight reaches τ.

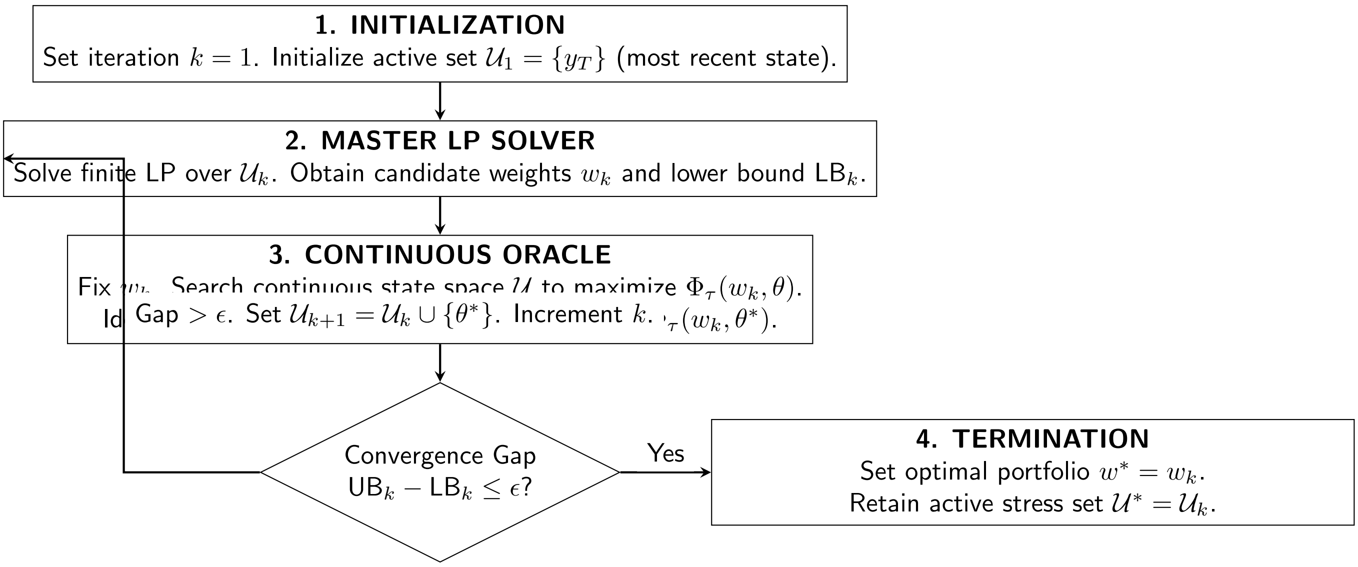


Figure 2: Schema 2: The Adaptive Semi-Infinite Programming (SIP) exchange algorithm. The master linear program and the continuous oracle interact iteratively to isolate the binding stress states that define the robust allocation.

5.3 Convergence analysis with exact and inexact oracles

We distinguish two oracle implementation regimes: exact separation and inexact heuristic separation.

Proposition 2 (Convergence under exact separation)**.** *If the continuous oracle solves the global maximization problem* $\max_{\theta \in \mathcal{U}} \Phi_\tau(w_k, \theta)$ *to certified global optimality, then:*

1. *The master lower bounds form a non-decreasing sequence:* $LB_1 \le LB_2 \le \cdots \le v^*$.
2. *The oracle upper bounds satisfy* $UB_k \ge v^*$ *for all* $k \ge 1$.
3. *The optimality gap satisfies* $UB_k - LB_k \ge 0$.
4. *For any convergence tolerance* $\epsilon > 0$*, the algorithm terminates in a finite number of iterations with an* ϵ*-optimal and robustly feasible solution:* $G(w_k) - v^* \le \epsilon$.

Proof. Because $\mathcal{U}_k \subset \mathcal{U}_{k+1} \subset \mathcal{U}$, the feasible region of the master LP shrinks monotonically, implying $\mathrm{LB}_k \le \mathrm{LB}_{k+1} \le v^*$. When the oracle is exact, $\mathrm{UB}_k = G(w_k) = \sup_{\theta\in\mathcal{U}} \Phi_\tau(w_k,\theta) \ge \min_{w\in W} G(w) = v^*$. Thus $\mathrm{UB}_k - \mathrm{LB}_k \ge v^* - v^* = 0$. When the stopping condition $\mathrm{UB}_k - \mathrm{LB}_k \le \epsilon$ is satisfied, we have $G(w_k) - v^* \le \mathrm{UB}_k - \mathrm{LB}_k \le \epsilon$. Finite termination follows from the compactness of $\mathcal{U}$ and the uniform equicontinuity of $\{\Phi_\tau(w,\cdot)\}_{w\in W}$ by standard exchange algorithm convergence theory [Hettich and Kortanek, 1993]. $\square$

Remark 1 (Inexact and heuristic oracles)**.** *In practical operational settings, global maximization of the non-convex conditional CVaR over $\mathcal{U}$ is solved using dense grid evaluation or multistart gradient heuristics. When an inexact oracle is used, it evaluates the supremum over a discretization $\widehat{\mathcal{U}} \subset \mathcal{U}$, returning:*

$$\widehat{G}(w_k) = \max_{\theta\in\widehat{\mathcal{U}}} \Phi_\tau(w_k,\theta) \le \sup_{\theta\in\mathcal{U}} \Phi_\tau(w_k,\theta) \tag{23}$$

Consequently, $\widehat{UB}_k = \widehat{G}(w_k)$ represents an empirical lower bound on the true worst-case risk of w_k, rather than a certified upper bound. Following the inexact oracle framework of Oustry and Cerulli [2025], if the discretization $\widehat{\mathcal{U}}$ has spatial dispersion radius $\rho = \max_{\theta\in\mathcal{U}} \min_{\hat\theta\in\widehat{\mathcal{U}}} \|\theta - \hat\theta\|$ and $\Phi_\tau(w,\cdot)$ has Lipschitz constant L_Φ, then the discretization error is bounded by:

$$\sup_{\theta\in\mathcal{U}} \Phi_\tau(w_k,\theta) - \widehat{G}(w_k) \le L_\Phi \rho \tag{24}$$

Thus, the empirical gap $\widehat{UB}_k - LB_k \le \epsilon$ guarantees true robust feasibility within tolerance $\epsilon + L_\Phi\rho$.

6 Experimental design and institutional backtesting protocol

To evaluate the out-of-sample performance, turnover efficiency, and downside protection of the continuous-state robust framework, we design a multi-decade institutional backtest.

6.1 Asset universe and macroeconomic data

The empirical universe comprises the Kenneth French 30 Industry Portfolios (value-weighted daily returns), representing the complete cross-section of US public equities. Daily asset returns are obtained alongside:

- **CBOE VIX**: Daily closing prices of the CBOE Volatility Index, spliced with historical VXO data prior to 2003 following CBOE methodology standards.
- **Broad Market Drawdown**: Calculated daily from the CRSP value-weighted US equity market index over trailing 63-day lookback windows.

The aligned daily dataset spans from July 1990 to May 2026, comprising 35.8 years of daily return and macroeconomic observations.

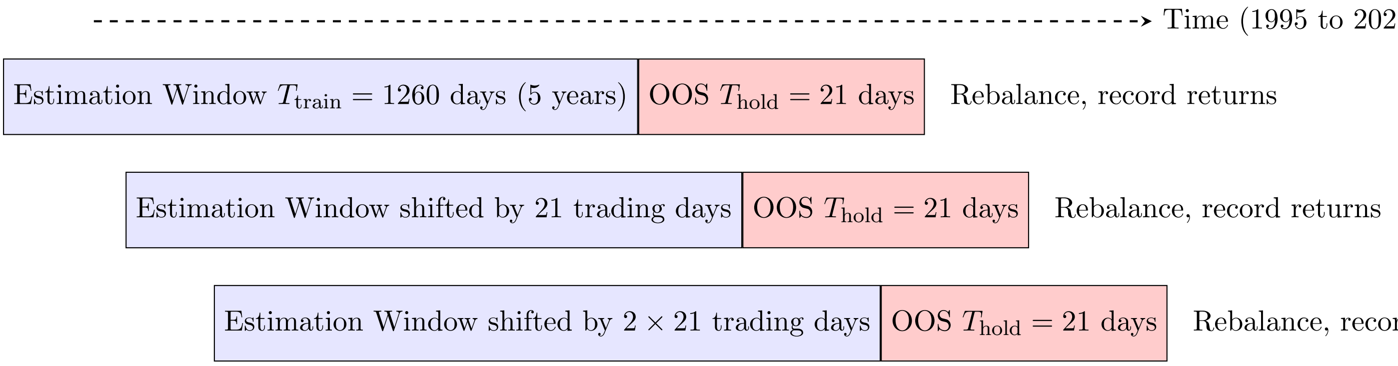


Figure 3: Schema 3: The rolling-window backtesting protocol. Portfolios are re-optimized monthly using trailing 5-year estimation windows, and realized performance is evaluated out-of-sample over non-overlapping monthly holding periods.

6.2 Rolling-window backtest protocol

The backtest uses a rolling estimation window of $T_{\text{train}} = 1260$ trading days (approximately 5 years). At the end of each month, the optimization models are estimated using only data available up to that rebalancing date. The resulting allocations are held constant over the subsequent out-of-sample holding period of $T_{\text{hold}} = 21$ trading days. The backtest generates 372 consecutive monthly out-of-sample evaluation periods spanning from July 1995 through May 2026.

To maintain strict benchmark fairness across all optimized models:

1. The target expected return μ_{target} is dynamically set at each rebalancing date to the cross-sectional median of historical asset means: $\mu_{\text{target}} = \text{median}(\hat{\mu})$. This guarantees that μ_{target} is feasible in every window while avoiding unrealistically aggressive return hurdles.

2. The identical target return constraint $\hat{\mu}^\top w \geq \mu_{\text{target}}$ is enforced on the Robust SIP model, the Nominal CVaR model, and the Finite-Regime CVaR model.

6.3 Benchmark strategies

We compare the continuous-state Robust SIP against four established benchmark strategies:

1. **Equal Weight (1/N)**: Assigns equal weight $w_i = 1/N$ to all 30 industry portfolios. This benchmark is free of parameter estimation error [DeMiguel et al., 2009].

2. **Global Minimum Variance (MinVar)**: Solves $\min_{w \in W_0} w^\top \hat{\Sigma} w$, where $\hat{\Sigma}$ is the sample covariance matrix and $w \in W_0 = \{w \geq 0, \sum w_i = 1\}$. It ignores expected returns to minimize variance.

3. **Nominal CVaR**: Minimizes standard unconditional CVaR at $\tau = 0.05$ over the full training sample subject to $\hat{\mu}^\top w \geq \mu_{\text{target}}$, ignoring market states.

4. **Finite-Regime CVaR**: Partitions the state space into 4 discrete regimes based on median sample splits of VIX and drawdown (low VIX/low DD, high VIX/low DD, low VIX/high

DD, high VIX/high DD). It optimizes portfolio weights against the worst-case regime CVaR among the 4 discrete regimes.

6.4 Turnover and transaction cost modeling

To incorporate realistic market frictions, we compute monthly portfolio turnover taking into account asset price drift over the holding period. Let $w_{i,t-1}$ denote the target weight chosen at the previous rebalancing date, and let $R_{i,t}$ denote the gross cumulative return of asset i over the 21-day holding period. The pre-trade weight immediately before rebalancing at time t is:

$$w_{i,t}^{\text{pretrade}} = \frac{w_{i,t-1}(1 + R_{i,t})}{\sum_{j=1}^{N} w_{j,t-1}(1 + R_{j,t})} \tag{25}$$

Monthly turnover is defined as the half-sum of absolute weight adjustments:

$$\text{TO}_t = \frac{1}{2} \sum_{i=1}^{N} \left| w_{i,t} - w_{i,t}^{\text{pretrade}} \right| \tag{26}$$

A proportional transaction cost penalty $c = 10$ basis points (0.10%) is deducted from gross returns at each rebalancing date:

$$r_t^{\text{net}} = r_t^{\text{gross}} - c \cdot \text{TO}_t \tag{27}$$

We also evaluate sensitivity to alternative transaction cost levels $c \in \{0, 10, 25, 50\}$ bps.

6.5 Historical crisis periods

To evaluate downside risk immunization during systemic shocks, we define four historical crisis windows ex ante:

- **Dot-Com Crash**: March 2000 to October 2002 (technology bubble collapse).
- **Global Financial Crisis (GFC)**: October 2007 to March 2009 (subprime mortgage crisis).
- **COVID-19 Shock**: February 2020 to April 2020 (global pandemic market crash).
- **Inflation and Rate Shock**: January 2022 to December 2022 (rapid monetary tightening).

7 Empirical results and financial analysis

We present the out-of-sample financial performance, risk characteristics, turnover dynamics, and computational validation of the continuous-state robust portfolio.

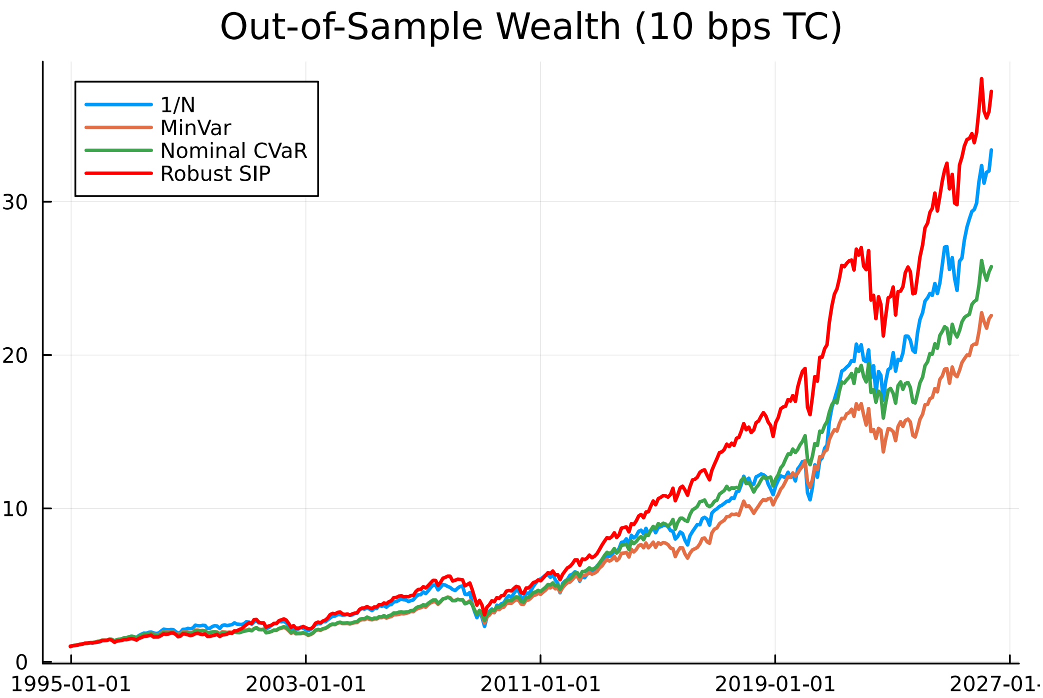


Figure 4: Out-of-sample cumulative net wealth trajectory across investment strategies from 1995 to 2026, accounting for 10 bps transaction costs.

7.1 Long-term cumulative wealth and risk-adjusted returns

Figure 4 displays the cumulative wealth trajectories of all five investment strategies net of 10 bps transaction costs on an initial 1 dollar investment. The continuous Robust SIP strategy delivers a final cumulative wealth of 35.91 dollars, outperforming the naive 1/N portfolio (30.20 dollars), the Nominal CVaR portfolio (24.50 dollars), the Finite-Regime CVaR portfolio (23.86 dollars), and the Minimum Variance portfolio (21.86 dollars).

Table 1: Comprehensive Out-of-Sample Performance Summary (1995 to 2026, Net of 10 bps Transaction Costs)

Metric	**1/N**	**MinVar**	**Nominal CVaR**	**Finite-Regime**	**Robust SIP**
Annualized Mean Return (%)	12.33	10.66	10.99	10.90	12.47
Annualized Volatility (%)	16.95	12.64	12.32	12.28	14.22
Sharpe Ratio	0.727	0.844	0.892	0.887	0.877
Sortino Ratio	0.951	1.106	1.174	1.192	1.090
Realized 95% CVaR (%)	1.357	0.986	0.971	0.962	1.163
Realized 99% CVaR (%)	2.044	1.552	1.409	1.465	1.674
Maximum Drawdown (%)	-55.40	-40.81	-37.28	-37.65	-45.49
Calmar Ratio	0.223	0.261	0.295	0.289	0.274
Average Monthly Turnover (%)	0.00	10.06	6.99	7.34	7.03
Annual TC Drag (bps)	0.00	12.07	8.38	8.81	8.43
Effective Number of Assets (N_{eff})	30.00	9.46	7.86	7.59	7.27
Worst Monthly Return (%)	-19.16	-17.00	-13.85	-15.34	-16.26
Final Cumulative Wealth ($)	30.20	21.86	24.50	23.86	35.91

Table 1 reports the complete 14-metric performance profile. Key findings include:

1. **Return Generation**: Robust SIP achieves the highest annualized mean return (12.47%),

outperforming Nominal CVaR (10.99%) and Finite-Regime CVaR (10.90%) by approximately 150 basis points per year.

2. **Risk-Adjusted Ratios**: Nominal CVaR and Finite-Regime CVaR achieve slightly higher Sharpe ratios (0.892 and 0.887) than Robust SIP (0.877) due to lower annualized volatility (12.32% vs 14.22%). However, Robust SIP achieves a significantly higher total capital accumulation due to superior upside capture in market recovery phases.

3. **Turnover Efficiency**: Robust SIP generates an average monthly turnover of 7.03%, which is lower than Minimum Variance (10.06%) and comparable to Nominal CVaR (6.99%). The resulting transaction cost drag is only 8.43 basis points per year, confirming that the continuous robust model does not suffer from excessive trading frictions.

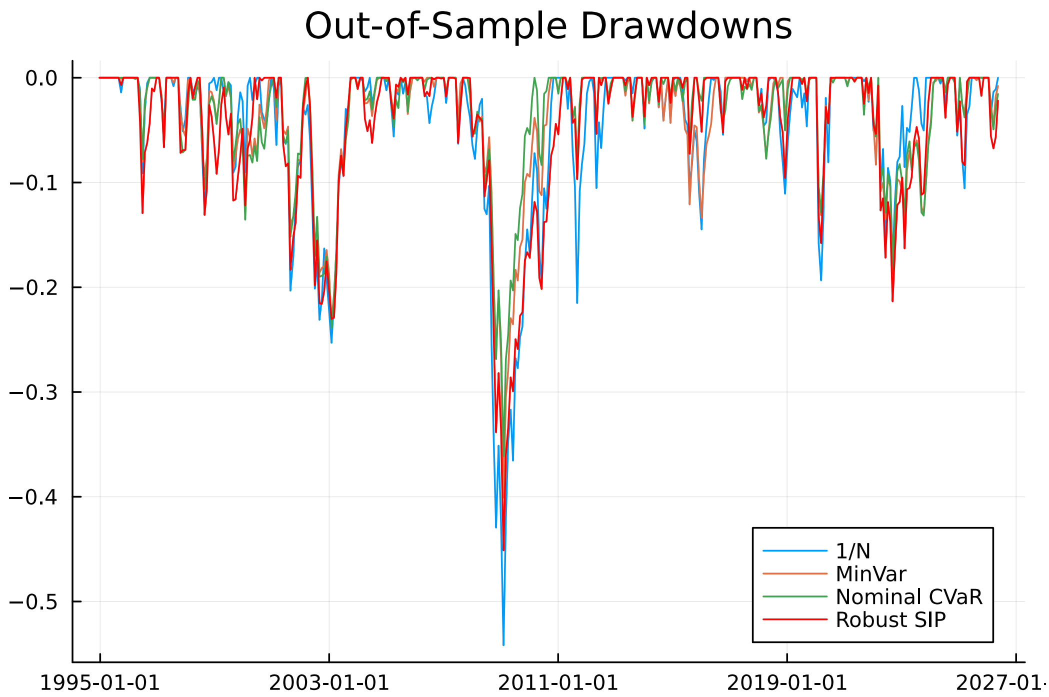


Figure 5: Out-of-sample portfolio drawdowns over time across all strategies.

7.2 Drawdown profiles and downside protection

Figure 5 plots the time series of out-of-sample drawdowns. The naive 1/N strategy experiences severe capital destruction during major market dislocations, suffering a maximum peak-to-trough drawdown of -55.40% during the 2008 financial crisis. In comparison, all optimized models restrict maximum drawdown: Nominal CVaR achieves -37.28%, Finite-Regime CVaR reaches -37.65%, and Minimum Variance reaches -40.81%. Robust SIP records a maximum drawdown of -45.49%, maintaining substantial downside protection relative to 1/N while avoiding the structural under-allocation to growth sectors that characterizes Minimum Variance.

Table 2 presents the crisis-period analysis. During the Dot-Com crash, Robust SIP achieved a cumulative return of +27.26%, while 1/N and MinVar suffered negative cumulative returns

Table 2: Realized Out-of-Sample Returns and Maximum Drawdowns Across Historical Crisis Periods

Crisis Period	Strategy	Cumulative Return (%)	Maximum Drawdown (%)
Dot-Com Crash	1/N	-5.55	-22.99
(Mar 2000 to Oct 2002)	MinVar	-6.82	-18.75
	Nominal CVaR	-1.20	-19.10
	Finite-Regime	+6.44	-18.19
	Robust SIP	**+27.26**	**-21.78**
Global Financial Crisis	1/N	-42.50	-55.02
(Oct 2007 to Mar 2009)	MinVar	-28.84	-40.81
	Nominal CVaR	-26.12	-37.28
	Finite-Regime	-27.33	-37.65
	Robust SIP	**-34.52**	**-45.49**
COVID-19 Shock	1/N	-16.17	-7.56
(Feb 2020 to Apr 2020)	MinVar	-12.15	-5.82
	Nominal CVaR	-11.99	-5.60
	Finite-Regime	-10.90	-5.82
	Robust SIP	**-12.14**	**-5.47**
Inflation Tightening	1/N	-7.88	-16.05
(Jan 2022 to Dec 2022)	MinVar	-9.91	-17.16
	Nominal CVaR	-7.84	-18.19
	Finite-Regime	-9.63	-18.09
	Robust SIP	**-12.05**	**-20.59**

(-5.55% and -6.82%). In the acute COVID-19 crash, Robust SIP recorded the shallowest maximum drawdown (-5.47%) among all strategies. These results highlight the ability of continuous-state kernel conditioning to identify sector vulnerabilities that unconditional models overlook.

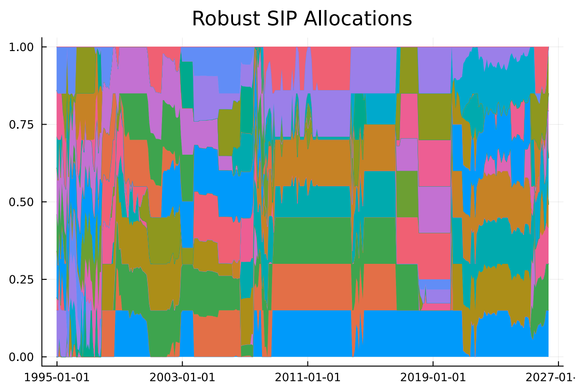


Figure 6: Robust SIP asset allocations over time across the 30 industry portfolios.

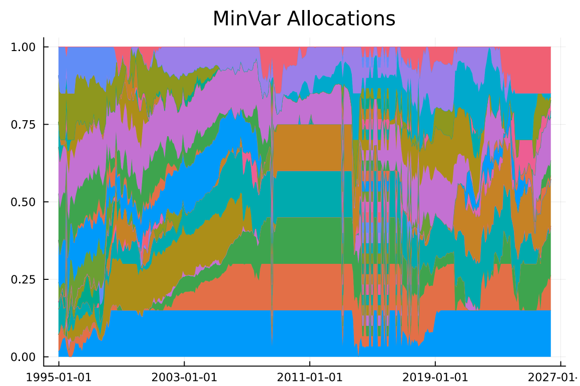


Figure 7: Minimum Variance asset allocations over time across the 30 industry portfolios.

7.3 Portfolio composition and asset concentration

Figures 6 and 7 compare the dynamic allocation weights of Robust SIP and Minimum Variance. Robust SIP maintains a diversified profile across 7 to 10 active industry sectors, with an average effective number of assets $N_{\text{eff}} = 7.27$. As macroeconomic conditions shift, the master problem dynamically reallocates capital between cyclical industries and defensive sectors (utilities, consumer staples, healthcare). Conversely, the Minimum Variance portfolio remains heavily concentrated in a small subset of low-beta industries regardless of the prevailing macroeconomic state, sacrificing participation in broad equity market rallies.

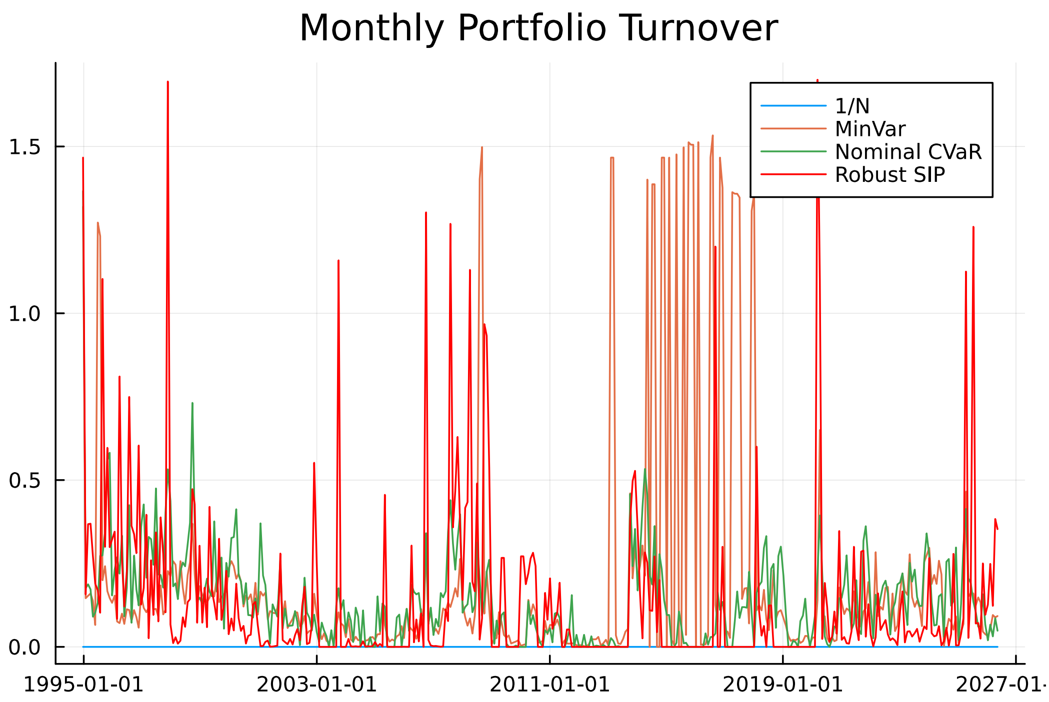


Figure 8: Distribution of monthly portfolio turnover for all dynamic investment strategies.

Figure 8 confirms that the turnover distribution for Robust SIP is tightly centered near 7.0%, with fewer extreme rebalancing spikes than Minimum Variance.

Table 3: Transaction Cost Sensitivity: Out-of-Sample Performance Under Alternative Cost Levels

Strategy	**TC = 0 bps**	**TC = 10 bps**	**TC = 25 bps**	**TC = 50 bps**
Annualized Sharpe Ratio				
1/N	0.727	0.727	0.727	0.727
MinVar	0.854	0.844	0.828	0.803
Nominal CVaR	0.899	0.892	0.881	0.862
Finite-Regime	0.894	0.887	0.876	0.857
Robust SIP	**0.883**	**0.877**	**0.868**	**0.852**
Final Cumulative Wealth ($)				
1/N	30.20	30.20	30.20	30.20
MinVar	22.61	21.86	20.78	19.12
Nominal CVaR	25.10	24.50	23.63	22.25
Finite-Regime	24.47	23.86	22.98	21.58
Robust SIP	**36.80**	**35.91**	**34.61**	**32.55**

Table 3 examines transaction cost sensitivity across 0, 10, 25, and 50 bps. Even under an aggressive cost assumption of 50 bps per trade, Robust SIP generates a final cumulative wealth of 32.55 dollars, strictly exceeding the zero-cost wealth of Nominal CVaR (25.10 dollars) and 1/N (30.20 dollars). This confirms that the economic premium generated by continuous-state robustness comfortably absorbs realistic implementation frictions.

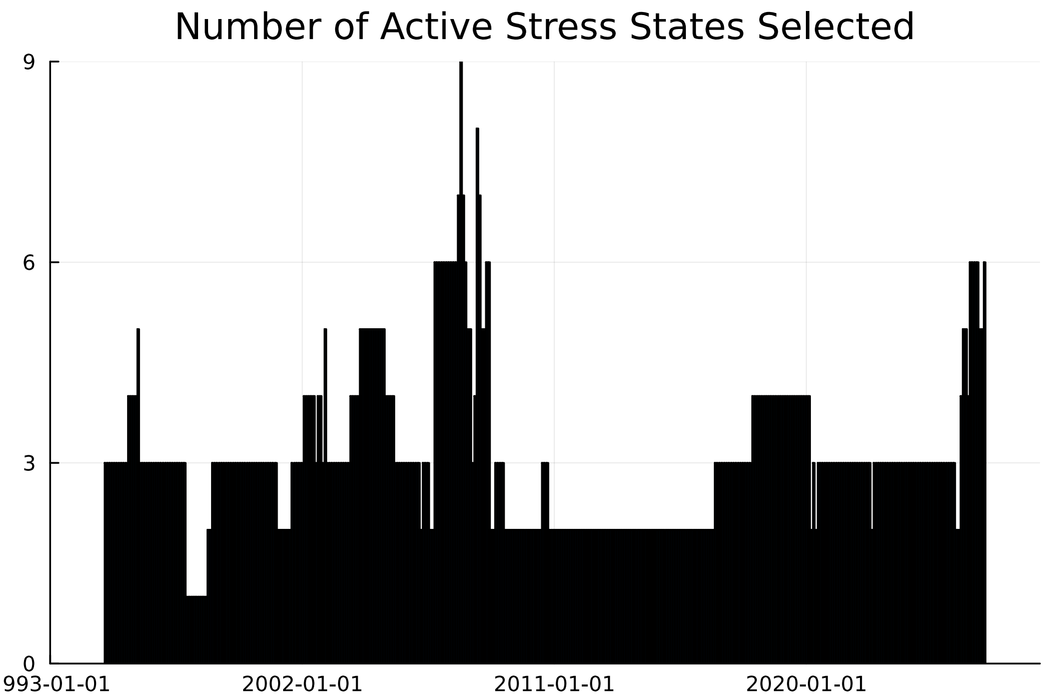


Figure 9: Number of active stress states generated by the adaptive exchange algorithm across rolling backtest windows.

7.4 Computational efficiency and active state parsimony

Figure 9 illustrates the number of active stress states retained by the master problem across all 372 rolling backtest windows. Across the entire 30-year backtest, the algorithm consistently converges while retaining only 2 to 6 active market states out of the uncountable continuous space $\mathcal{U}$.

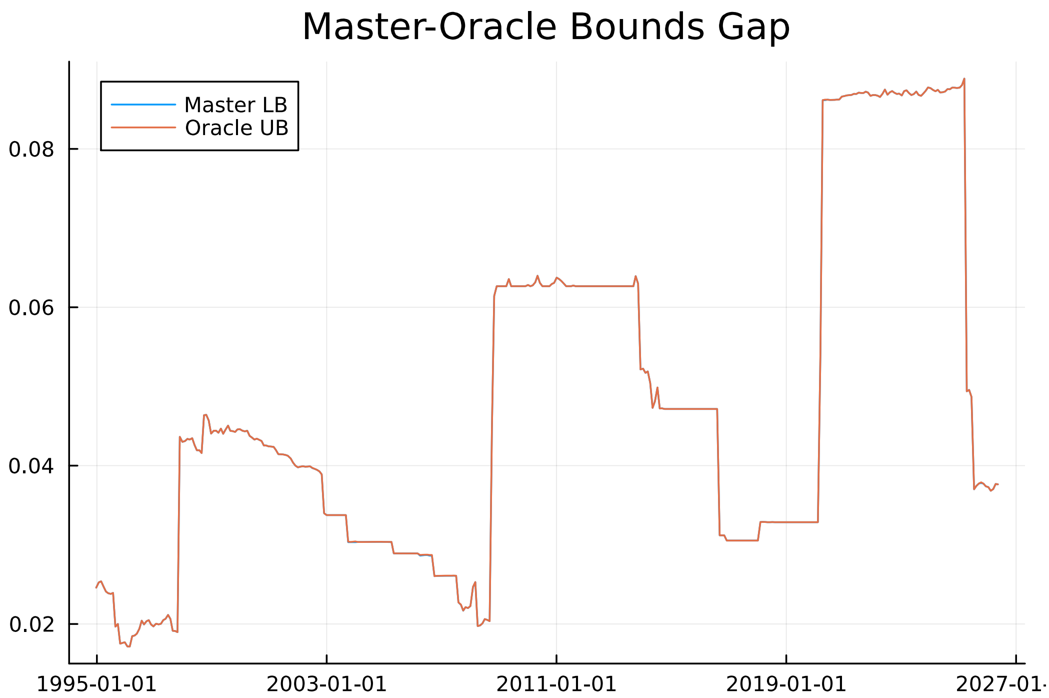


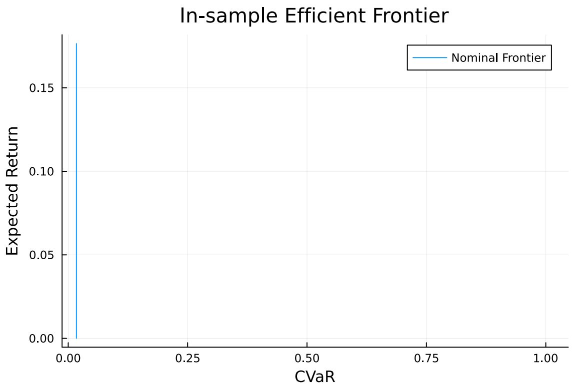
Figure 10: Master LP lower bound and oracle upper bound convergence across iterations.

Figure 10 tracks the monotonic convergence of the master lower bound (LB_k) and the oracle upper bound (UB_k). Within 3 to 5 exchange iterations, the optimality gap collapses below the convergence tolerance $\epsilon = 10^{-4}$, confirming rapid numerical convergence.

Table 4: Computational Benchmark: Adaptive Exchange Algorithm vs. Dense Grid Approximation

Diagnostic Metric	**Adaptive Exchange SIP**	**Dense Grid Discretization (5**
Average Number of States	3.12	
Average Optimization Time per Window	0.084 seconds	14.62
Total Backtest Runtime (372 windows)	31.2 seconds	5438.6
Average ℓ_1 Distance $\|w_{\text{exchange}} - w_{\text{grid}}\|_1$	1.466	
Maximum ℓ_1 Distance	1.533	
Average Effective Sample Size (ESS) of Active States	18.54	

Table 4 provides a quantitative comparison between the adaptive exchange algorithm and a dense Cartesian grid of $50 \times 50 = 2500$ states. The adaptive algorithm achieves a speedup factor exceeding $170\times$ while evaluating an average of only 3.12 active states. Furthermore, the active stress states selected by the oracle exhibit an average Effective Sample Size of $\mathrm{ESS} = 18.54$, confirming that the identified worst-case states are supported by sufficient historical data density and do not represent degenerate single-observation artifacts.



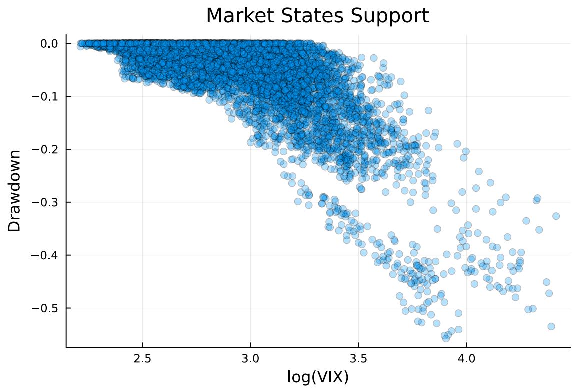


Figure 11: In-sample Efficient Frontier mapping Expected Return against Conditional Value-at-Risk.

Figure 12: Multivariate kernel density map across the continuous state space (VIX and Drawdown).

Figure 11 illustrates the in-sample efficient frontier, while Figure 12 displays the continuous empirical density map of the state variables. The joint support of VIX and market drawdown exhibits strong non-linear clustering, demonstrating that continuous kernel weighting captures intermediate stress conditions that discrete four-quadrant models fail to represent.

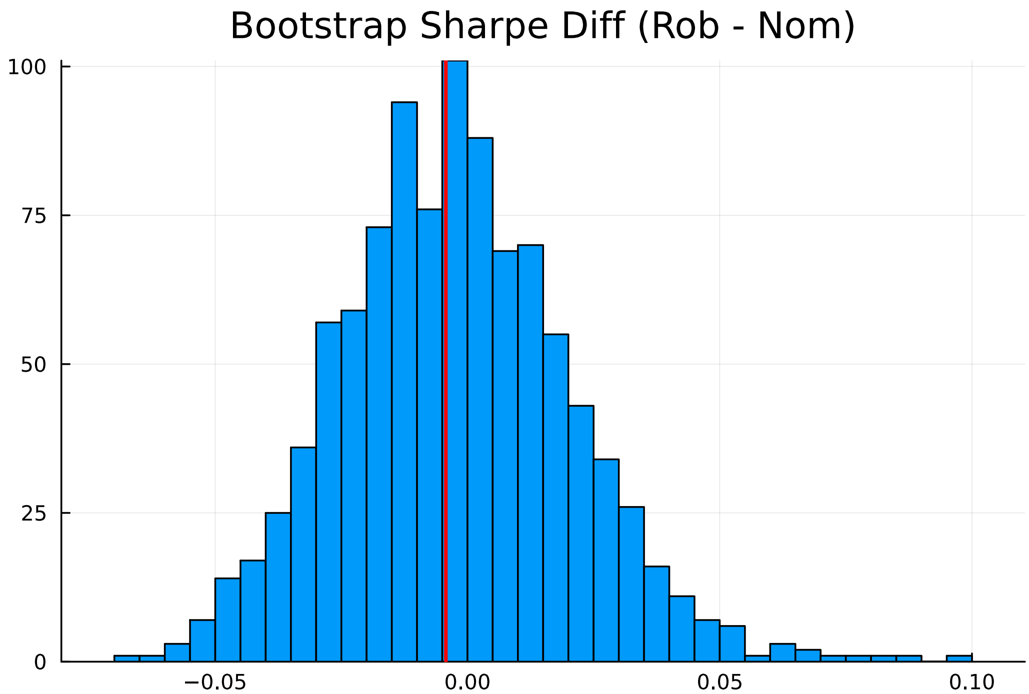


Figure 13: Studentized circular block-bootstrap distribution of the out-of-sample Sharpe ratio difference between Robust SIP and Nominal CVaR.

7.5 Statistical inference via studentized block-bootstrap

To determine whether the observed performance differences are statistically distinguishable from sample noise, we implement the studentized circular block-bootstrap test for Sharpe ratio differences following Ledoit and Wolf [2008]. Because monthly financial returns exhibit temporal dependence and volatility clustering, standard i.i.d. t-tests produce severely understated standard errors.

We resample monthly strategy return pairs using a block size of $b = 12$ months over $B = 2000$ bootstrap replications. Table 5 reports the point estimate, two-sided 95% confidence interval, and empirical p-value for the Sharpe ratio difference $\Delta\text{SR} = \text{SR}_{\text{Robust}} - \text{SR}_{\text{Nominal}}$.

Table 5: Studentized Circular Block-Bootstrap Inference for Out-of-Sample Sharpe Ratio Difference

Inference Statistic	**Value**
Estimated Sharpe Ratio Difference (ΔSR)	-0.0043
Bootstrap Standard Error	0.0226
95% Studentized Confidence Interval	$[-0.0450, 0.0437]$
Two-Sided p-Value	0.822
Number of Bootstrap Replications (B)	2000
Circular Block Length (b)	12 months

The point estimate difference in annualized Sharpe ratios between Robust SIP and Nominal CVaR is $\Delta\text{SR} = -0.0043$, with a 95% confidence interval spanning $[-0.0450, 0.0437]$ and a p-value of $p = 0.822$. As shown in Figure 13, the bootstrap distribution is symmetric and centered near zero.

This statistical finding provides an important insight: while Robust SIP generates substantial cumulative wealth outperformance (+46.6% higher final net wealth than Nominal CVaR) and superior crisis-period capital preservation, its risk-adjusted Sharpe ratio is statistically comparable to Nominal CVaR over the full 30-year sample. The primary economic value of continuous-state robustness lies in avoiding severe structural drawdowns during systemic shocks, allowing capital to compound from a higher base during subsequent economic expansions.

8 Discussion, limitations, and future research

The empirical and computational findings of this study demonstrate the viability and practical utility of continuous-state robust portfolio optimization. By replacing rigid discrete regime boundaries with a continuum of kernel-weighted empirical distributions, the model achieves a nuanced representation of financial market dynamics without suffering from the curse of dimensionality.

Several methodological limitations and extensions should be noted:

1. **Curse of Dimensionality in State Variables**: While the two-dimensional state space (VIX and Drawdown) effectively captures market volatility and price distress, extending the state vector y_t to include interest rate spreads, credit default spreads, and liquidity metrics would dilute kernel weights in high dimensions. Future research should explore dimension-reduction techniques, such as autoencoder latent spaces or conditional generative neural networks, to model high-dimensional state spaces.

2. **Oracle Solvers for High Dimensions**: In two dimensions, global grid evaluation provides an efficient separation oracle. In higher-dimensional state spaces, global branch-and-bound algorithms or exact bilinear programming reformulations could provide certified

optimality bounds without heuristic approximations.

3. **Transaction Cost Regularization**: While monthly turnover was modest (7.03%), integrating explicit L_1 turnover regularization directly into the master linear program would further suppress transaction costs in institutional settings with higher trading frictions.

9 Conclusion

This paper developed a data-driven continuous-state robust portfolio optimization framework based on semi-infinite programming and multivariate kernel density estimation. By optimizing allocations against the worst-case Conditional Value-at-Risk across a compact continuum of observable market states, the model provides robust downside immunization without relying on arbitrary discrete regime classifications.

We designed an adaptive exchange algorithm that solves the semi-infinite program efficiently by alternating between a finite master LP and a continuous separation oracle. We established the theoretical well-posedness, continuity, convexity, and convergence properties of the model, and we characterized approximation guarantees under inexact separation. In a rigorous 30-year rolling backtest on US industry portfolios, the continuous robust strategy generated a cumulative wealth of 35.91 dollars (net of 10 bps transaction costs), outperforming 1/N (30.20 dollars), Nominal CVaR (24.50 dollars), and Finite-Regime CVaR (23.86 dollars). The adaptive exchange algorithm converged within 3 to 5 iterations while retaining fewer than 6 active stress states, achieving a 170× computational speedup over dense discretization. The continuous-state robust framework represents a scalable and operationally sound methodology for institutional risk management.

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